

SwadeshAI: Building India’s AI Sovereignty through Fundamental Research

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Abstract

India is the world’s largest mobile user base for ChatGPT and the third-largest for DeepSeek, yet contributes between 0.1 and 1% of accepted papers at top artificial intelligence (AI) conferences and has produced no frontier-class AI architectures or foundation models. The Stanford AI Index records India at roughly 8–9% of global AI publications, yet India remains largely absent from the rankings of notable models and high-impact AI research. India’s Rs.10,372 Cr IndiaAI Mission allocates nothing to fundamental AI research in theory, algorithms, or mathematical foundations. This paper argues that AI sovereignty—the capacity to build, deploy, and govern indigenous AI capabilities—cannot be achieved through an application-and-services strategy alone. Drawing on bibliometric evidence from the Stanford AI Index, Scopus, and the IndiaML tracker, as well as comparative analysis of China’s trajectory from AI follower to near-parity with the United States, we show that foundational research capability is the binding constraint on India’s AI self-reliance story. To remedy this situation, we propose a set of policy interventions: earmarking a dedicated AI research fund within the Anusandhan National Research Foundation (ANRF), reforming UGC and AICTE promotion criteria from publication count to research quality, and creating a small number of autonomous, privately funded or public–private frontier research labs focused on high-risk, long-horizon AI research. We further argue that India should build capability not only in current foundation-model technologies, but also in two or three plausible next-generation directions where a future technological disruption may create new opportunities for leadership. Together, these measures can expand India’s frontier research capacity while reducing the risk of remaining permanently in catch-up mode.

Keywords: Atmanirbhar Bharat; Sovereign AI; artificial intelligence; R&D policy; fundamental research

1. Introduction

India’s relationship with artificial intelligence (AI) is defined by a striking paradox. The country is the world’s largest market for ChatGPT’s mobile application and the third-largest user base for DeepSeek [20]. India’s AI market is projected to reach \$8 billion by 2025, growing at 40% compound annual growth rate [19]. Indic-language AI represents a massive latent opportunity. India constitutes 18% of the global population, yet Indic languages comprise merely 1% of the

Common Crawl corpora [10] that is extensively used in training frontier large language models (LLMs). The demographic and market opportunity for AI leadership could hardly be more favourable, and the moment for decisive action has arrived.

Data curated by the Stanford AI Index [19, 20] paint a bleak picture of AI R&D in India. India produced zero of the “notable AI models” tracked by the Stanford AI Index, while the United States produced 59 and China 35 [20, Figure 1.1.1]. India contributes between 0.1 and 1% of accepted papers at leading AI conferences [2, 9]. India’s share of global AI patents stands at 0.4%, compared to China’s 74.2% and the United States’ 12.1% [20, Figure 1.7.2]. Among the list of countries contributing to the 100 most-cited AI publications in 2021–2024, India is at the last place with 1-4 publications per year, with small nations like Hong Kong, Switzerland, and Singapore appearing higher [20, Figure 1.6.12].

This distinction is particularly important because technological leadership is rarely static. AI itself has already undergone multiple paradigm shifts: from symbolic and knowledge-based systems, through statistical and probabilistic methods, to neural networks and today’s foundation-model era. The relevant policy question is therefore not simply whether India can reproduce today’s frontier technologies, but whether it can build the research capability to participate when the frontier changes again. A country that begins investing only after a new paradigm has matured is likely to remain a follower; a country with a sufficiently deep research base can begin experimenting before the direction of technological change is clear. For this reason, an AI sovereignty strategy should combine capability-building in current technologies with a small portfolio of bets on plausible next-generation approaches.

The objective of sovereign AI should therefore not be interpreted as not just reproducing the AI stack as it is already developed elsewhere, but *also* developing enough indigenous scientific capability to shape the next technological frontier. However, one can argue that the *swadeshi* aspiration embedded in the *Atmanirbhar Bharat* vision—building at home rather than importing—is precisely what is absent in India’s current AI strategy, which imports the intellectual foundations and indigenises only the surface layer. India’s Rs.10,372 Cr IndiaAI Mission reflects this orientation. It devotes the overwhelming majority of its budget to compute infrastructure, startup financing, and application development, with no explicit allocation for curiosity-driven basic research from which sovereign capability is built [5, 13].

The paper is structured as follows. Section 2 presents bibliometric evidence mapping India’s AI research deficit. Section 3 diagnoses the structural root causes. Section 4 examines China’s trajectory as a comparative counterfactual, and draws on South Korea’s success story of R&D-driven growth. Section 5 proposes policy recommendations to make progress on our broad vision, and Section 6 offers a conclusive discussion.

2. Mapping the Deficit: India's AI Research in Global Context

Drawing on the Stanford AI Index reports for 2025 [19] and 2026 [20], we assess India's position in the global AI research landscape along three dimensions: the volume and impact of its scholarly publications, its representation at top AI research conferences, and its capacity to produce frontier AI models. Going beyond large language models (LLMs), these indicators capture the depth of the research ecosystem that produces new methods, architectures, algorithms, and scientific understanding. Across all three, the same pattern emerges: impressive surface-level metrics that dissolve on contact with quality-adjusted comparisons.

2.1. Volume without impact

India's research output in AI, measured by publication volume, appears respectable (Figure 1 left panel). India accounted for 9.22% of global AI publications in 2023, roughly on par with the United States at 9.2%, though well behind China at 23.2% [19]. In the broader Scopus database, Indian researchers contributed more than 16 Lakh papers between January 2020 and May 2025: the highest volume in the world [1].

However, this volume masks a deeper quality deficit. When the impact of published papers in AI is considered, measured in citations generated, India trails substantially (Figure 1 right panel). Chinese AI publications accounted for 22.6% of all AI citations in 2023, followed by Europe at 20.9% and the United States at 13.0%; India's share was 6.1% [19]. India's H-index of 925 is 71.2% lower than the United States' H-index of 3,213 [1]. Of the roughly 200 Indian journals indexed in Scopus, only 12 are classified as Quartile 1 (top 25%), with the vast majority sitting in Quartiles 3 and 4 [1].

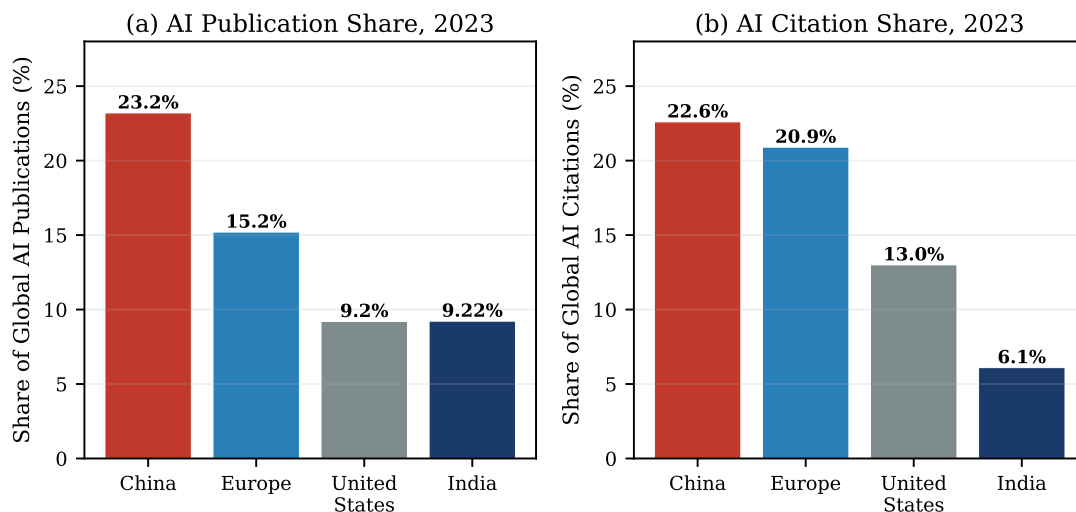


Figure 1: AI publication versus citation shares by country/region, 2023. *Source: Stanford HAI [19], Figures 1.1.6–1.1.7.*

Looking at how things changed over time, between 2013 and 2023, China's publication share rose from approximately 10% to 23.2% whilst its citation share grew commensurately from 8%

to 22.6%. In comparison, India’s publication share grew from approximately 5% to 9.2% over the same period, but its citation share grew far more slowly, from approximately 3% to 6.1% [19]. India’s papers are produced at increasing volume but are not read, cited, or built upon at a commensurate rate. This indicates a research ecosystem that is optimised for production rather than influence.

2.2. The top-conference gap

In AI, unlike most scientific disciplines, peer-reviewed conferences rather than journals serve as the primary venue for disseminating frontier research. The IndiaML tracker, an open-source initiative tracking Indian-affiliated papers at major machine learning venues using the OpenReview API, documents that India’s contribution to accepted papers at four top AI conferences (NeurIPS, ICML, ICLR, and CVPR) ranges between 0.1 and 1% [2, 9]. This is an extraordinarily low figure for a country with the world’s largest developer workforce and the largest consumer market for AI platforms.

The most revealing metric, however, is the distribution of the 100 most-cited AI publications. In 2024, the United States contributed 46 papers, China 41, Hong Kong 15, Australia 14, and the United Kingdom 9. India contributed 4. This pattern holds across the previous three years for which data are available. Overall, India lies far behind others in top-100 AI paper counts across 2021–2024 [19]. Figure 2 presents this data. The implications extend well beyond bibliometric standing. Top-cited papers introduce the architectures, training methods, and theoretical results that subsequent work builds upon; they define what the field studies and what counts as progress. A country absent from this list cedes the power to set that direction to others. Moreover, when India’s AI researchers work on problems framed by American and Chinese researchers, using methods designed for deployment contexts quite different from India’s own, the result is a body of work that solves the wrong problems at scale. As a consequence, the unique challenges Indian AI users face, such as lack of network connectivity, availability on-the-edge, and multilingual and code-switched usage remain unaddressed or suboptimally addressed.

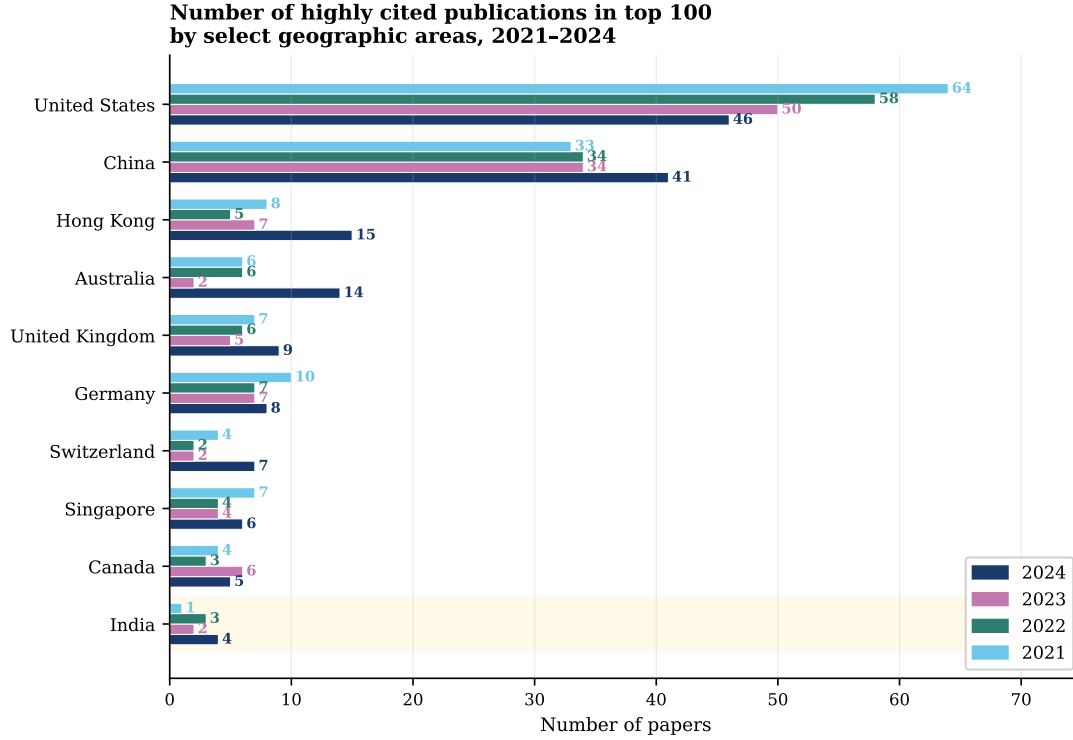


Figure 2: Number of papers in the top-100 most-cited AI publications, by country, 2021–2024. *Source:* Stanford HAI [20, Fig. 1.6.12].

2.3. The frontier-models gap

The Stanford AI Index tracks “notable AI models” representing significant technical advances. In 2024, the United States produced 40 such models, China 15, and Europe 3; in 2025, the United States produced 59 and China 35. India produced zero in both years [19, 20]. Indian foundation models do exist: Krutrim (Ola, trained on 2 trillion tokens across 10+ Indian languages), the Param series (BharatGen/IIT Bombay consortium, up to 17 billion parameters in 22 languages), and the Sarvam models launched under the India AI Impact Summit 2026 (30B and 105B parameter variants pretrained from scratch on large Indic corpora). Among them, the Sarvam models are the most substantial contribution, since pretraining from scratch on Indic data requires significant compute and data-curation effort, and signals a maturing of indigenous model-building capacity [10]. Yet even here, the ceiling is clear: none of these models appear on global leaderboards, and none has produced methodological innovations such a new architecture, training objective, or algorithmic technique comparable to those emerging from American and Chinese laboratories (a point we develop in Section 4). Building a pretrained Indic model is a necessary step toward AI sovereignty; it is not the same as advancing the science of building and understanding AI models. A nation with genuine AI sovereignty should ideally progress both. Figure 3 illustrates this frontier-models gap, and Table 1 summarises the full consumer-creator asymmetry.

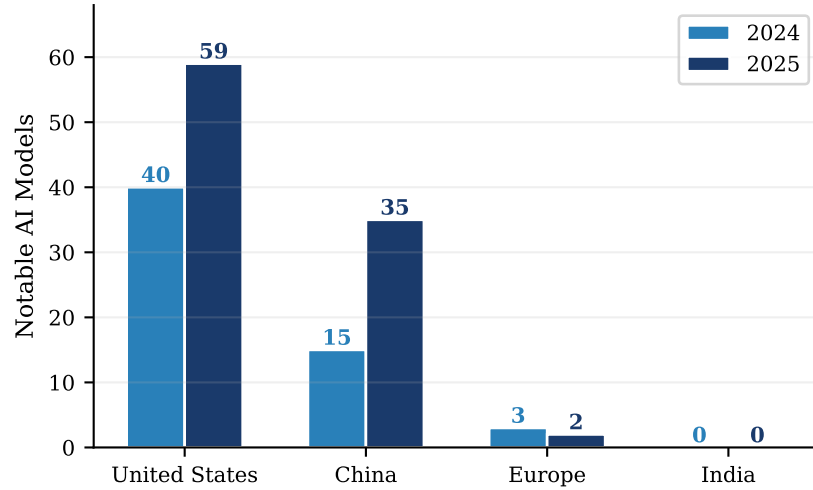


Figure 3: Notable AI models produced by country of origin, 2024–2025. India produced zero notable models in both years. AI sovereignty requires the capacity to produce frontier models, not merely to localise them. *Source:* Stanford HAI [19, 20].

Table 1: India’s AI consumer–creator paradox: key metrics across dimensions of AI sovereignty.

Metric	India	China	United States
AI publications (% global, 2023)	9.22%	23.20%	9.20%
AI citations (% global, 2023)	6.10%	22.60%	13.03%
Top-100 cited AI papers (2024)	4	41	46
Notable AI models (2025)	0	35	59
AI patents (% world total, 2023)	0.4%	74.24%	12.06%
Top-conference papers (% accepted)	0.1–1%	~30–40%	~30–40%
Private AI investment (2025)	\$1.25B (govt)	\$12.4B	\$285.9B
GERD/GDP	0.64%	2.68%	3.59%
Frontier foundation models	None	DeepSeek-R1, Qwen	GPT-4o, Claude
ChatGPT mobile user base	Largest	—	—
DeepSeek user base	3rd largest	1st	—
Semiconductor design workforce	20%	—	—
Semiconductor fabrication capacity	0.5%	—	—

Sources: Basak and Chopra [2], Endiya Partners [6], Indian Society of Artificial Intelligence and Law [9], Kallappa et al. [10], Ministry of Electronics and Information Technology [13], Organisation for Economic Co-operation and Development [17], Stanford HAI [19, 20]. Bold entries highlight the sharpest dimensions of the consumer-creator asymmetry.

3. Diagnosing the Root Cause

Four mutually reinforcing factors explain why India’s AI ecosystem has taken the shape it has: the composition of its flagship national AI investment, a misaligned academic incentive system,

a shortage and concentration of frontier research talent, and a deeply entrenched services-over-research orientation in the private sector. Each reinforces the others, and addressing any one of them in isolation will be insufficient without simultaneous reform of the wider research ecosystem.

3.1. The IndiaAI Mission’s architecture

The IndiaAI Mission, approved by the Cabinet in March 2024 with a five-year budget of Rs.10,372 Cr (approximately USD \$1.25 Billion), is India’s flagship national AI initiative [13]. Its seven pillars and their allocations are: Compute Capacity (Rs.4,563 Cr, 44%), Innovation Centre (Rs.1,971 Cr, 19%), Startup Financing (Rs.1,943 Cr, 18.7%), Future Skills (Rs.883 Cr, 8.5%), Application Development (Rs.689 Cr, 6.6%), Datasets Platform (Rs.200 Cr, 1.9%), and Safe & Trusted AI (Rs.20 Cr, 0.2%) [5] (Figure 4).

The budget reveals a structural bias towards deployment over discovery. Nearly two-thirds of the allocation funds hardware and capital access. The Innovation Centre¹—the pillar supposedly oriented towards research—is tasked with developing “indigenous Large Multimodal Models and domain-specific models” [13], which in practice means applied model-building, not fundamental research in learning theory, optimisation and algorithms, or statistical machine learning. Importantly, no pillar is explicitly dedicated to curiosity-driven basic AI research.

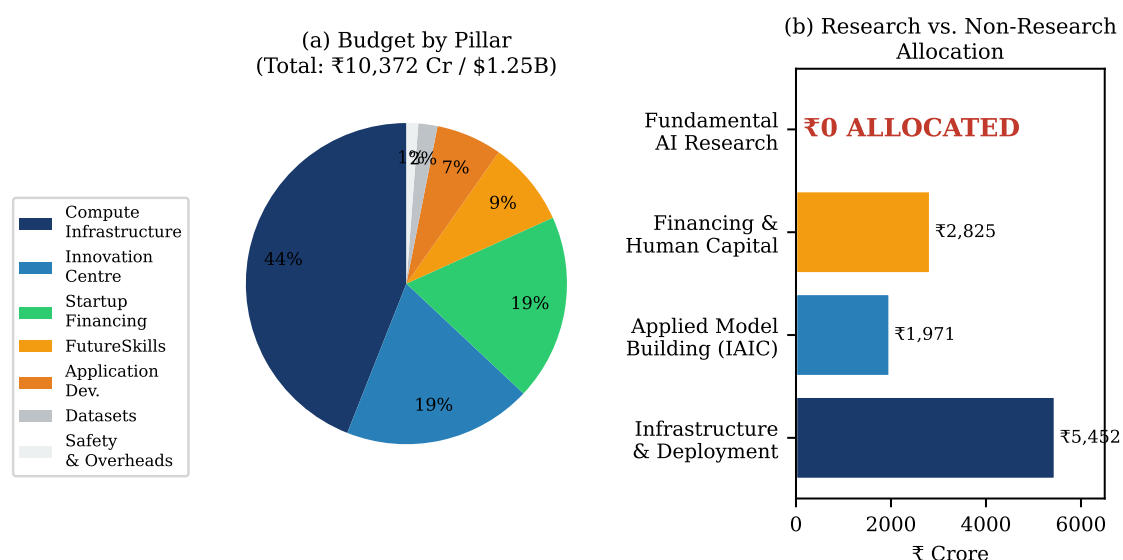


Figure 4: IndiaAI Mission budget breakdown by pillar (left) and reframed as research versus non-research allocation (right). No funding is explicitly allocated to fundamental AI research. A mission structured entirely around deployment, applied model-building, and financing cannot deliver AI sovereignty: it produces AI users, not AI inventors. *Source:* Parliamentary response, December 2024 [5]; Cabinet approval [13].

A sovereignty agenda requires investment commensurate with external benchmarks in similar category. Looking from this comparative perspective compounds the above structural concern.

¹<https://indiaai.gov.in/hub/indiaai-innovation-centre>

The IndiaAI budget of USD \$1.25 billion over five years translates to approximately USD \$250 million per year. In comparison, private investment in AI in the United States in 2025 alone was USD \$285.9 billion, roughly 1,140 times India’s annual government allocation [20]. Even within Asia, China’s government semiconductor fund alone is \$47.5 billion, 38 times India’s entire five-year AI budget [19].

3.2. Misaligned incentives

India’s academic promotion criteria, governed by the University Grants Commission (UGC) and All India Council for Technical Education (AICTE) regulations, reward publication count in Scopus- or Web of Science-indexed outlets rather than the quality of the publication venue or citation impact. This creates a rational incentive to publish in Q3 and Q4 journals, where acceptance is easier, rather than attempt the high-risk, high-reward submissions in world-class venues with much lower acceptance rates. Between 2020 and 2025, India contributed 1.6 million Scopus papers, the most in the world, but only 12 Indian journals achieve Q1 status [1]. A 2025 study of the top 1% of highly cited India-affiliated papers found that papers with more than 90% self-citations are predominantly domestic, small-team, and recent—suggesting that citation inflation, rather than genuine impact, is being incentivised [11].

This misalignment has a direct consequence for AI research talent. A faculty member at an Indian institution faces no career incentive to spend two years developing a theoretically novel contribution and submitting to a top conference like NeurIPS, where rejection is probable and the timeline uncertain, when five Q4 journal papers produce the same promotion outcome in the same period. The problem is compounded by the concentration of India’s existing frontier research capacity. An analysis of India’s contribution to leading AI conferences found that approximately 70% of India’s contribution comes from just 8 institutions, while roughly 90% of India’s AI papers come from around 20 institutions [3]. This concentration matters for policy: incremental funding spread uniformly across the research system is unlikely to produce frontier capability at the scale required. A more plausible strategy is to build a small number of exceptionally strong research clusters while simultaneously expanding the pipeline of PhD students, postdoctoral researchers, and industry–academic scientists.

3.3. The IT-services comfort zone

India’s technology ecosystem, built around major consulting and services companies, is fundamentally configured for execution rather than invention. These organizations are designed and built around the risk-averse strategy of simply deploying already mature technologies without investing in forward-looking research. As a contextual example in the domain of AI, they are consumers of open-source libraries like PyTorch and TensorFlow, without contributing substantially to either the algorithmic foundations or their worldwide development efforts. The corporate R&D culture exemplified by global success stories such as AT&T Bell Labs and IBM Research in the USA, Huawei Noah Lab and ByteDance’s AI Lab in China, or Samsung

Advanced Institute of Technology (SAIT) in Korea has no Indian equivalent at scale.

This pattern of downstream capability building without upstream research is self-reinforcing and structurally inimical to AI sovereignty. Indian IT firms, having built profitable businesses around service delivery, face no competitive pressure to invest in fundamental research. Their clients—predominantly American and European firms—perform the upstream R&D themselves. As AI increasingly commoditises the deployment layer through APIs, pretrained models, and low-code automation frameworks, the value will shift further upstream to those who create the foundational technologies. India’s current positioning leaves it exposed to precisely this shift.

The semiconductor story illustrates the stakes. India accounts for 20% of the global semiconductor design workforce, approximately 1.5 Lakh professionals, but contributes only 0.5% to global fabrication capacity [6]. Design-without-fabrication is the hardware-layer analogue of India’s AI situation: downstream capability without the sovereign control that comes only from owning the foundational layer.

4. Success Stories: How Fundamental Research can Build Self-reliance

Recent and relevant progress in two countries, China and South Korea, offer instructive counterpoints to India’s trajectory. Both began from positions of technological followership broadly comparable to India’s today, and both made deliberate, sustained investments in basic research as a matter of explicit national policy. Their outcomes demonstrate that the pipeline from foundational research to applied capability to sovereign competitiveness is real, reproducible, and historically achievable within a generation.

4.1. China

China’s transformation from a follower to a near-peer of the United States in broad R&D provides the most relevant comparative lesson for India (Figure 5). In 1996, China’s gross expenditure on R&D (GERD) as a percentage of GDP was 0.56%, comparable to India’s 0.76% at the time [21]. By 2023, China had nearly quintupled this ratio to 2.58%, with the National Bureau of Statistics reporting a further increase to 2.68% by 2024 [14]; India’s ratio, meanwhile, peaked at 0.86% in 2008 before declining, with the most recent official figure standing at 0.64% for 2020–21 and parliamentary responses indicating it has remained near 0.7% since [4, 18]. China’s total R&D expenditure in 2024 exceeded 3.6 trillion yuan (approximately USD \$500 billion), and on a purchasing-power-parity basis China surpassed the United States for the first time in absolute R&D spending, at USD \$1.03 trillion versus the USA’s USD \$1.01 trillion [12, 17].

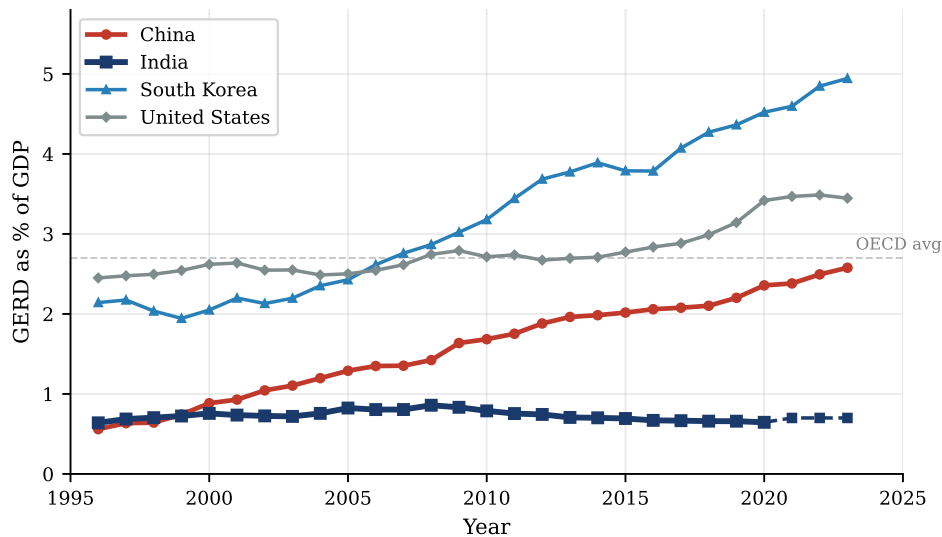


Figure 5: Gross expenditure on R&D as a percentage of GDP, 1996–2023. China rose substantially from 0.56% to 2.58% whilst India’s trajectory, which peaked at 0.86% in 2008, declined to 0.65% by 2020. South Korea, at 4.94%, leads the world. The divergence between China and India exemplifies two nations that started from the same point and made fundamentally different choices about research investment: choices whose consequences are now visible in AI sovereignty outcomes. India’s dashed segment (2021–2023) is carried forward at 0.7%, the upper bound cited in parliamentary responses, as DST has not yet published official GERD data beyond 2020–21. *Sources:* Department of Science and Technology [4], National Bureau of Statistics of China [14], Organisation for Economic Co-operation and Development [17], Press Information Bureau [18], World Bank [21].

In AI research outputs, China went from under 5% of global AI publications in 2000 to 36% by 2025, emerging as the single most dominant contributor [8]. China now leads in both AI publication volume (23.2% in 2023) and citation share (22.6%), and has recently overtaken even in high-impact publications, challenging the assumption that Western powers retain dominance in influential AI scholarship [8, 19]. In comparison, analysis of accepted papers at NeurIPS, ICML, and ICLR using the IndiaML tracker consistently places Chinese-affiliated authors among the top contributors by country, whilst Indian-affiliated authors account for under 1% of accepted papers [2, 9].

China’s AI research build-up was embedded in a broader basic-research expansion. In 2024, China’s investment in basic research grew 10.7% year-on-year to 250.1 billion Yuan, representing 6.9% of total R&D expenditure [15]. University capacity massively expanded in parallel: the number of higher education institutions grew from approximately 1,000 to over 3,000 between 2000 and 2024. Today, Chinese institutions hold 7 of the top 10 positions in the Nature Index [16]. The Chinese Academy of Sciences alone, with a Nature Index Share of 2,243, outproduces all of India combined, which has a nature index share of approximately 1,494 [16].

The DeepSeek story distils this lesson. A relatively small Chinese team, operating under US chip export controls that restricted access to advanced GPUs, produced DeepSeek R1, an open-source LLM that matched frontier American proprietary systems. This was possible because the

algorithmic innovation behind DeepSeek R1—an efficient neural network architecture and a novel reinforcement learning algorithm—did not require frontier hardware. Instead, it was made possible by the investment on frontier research [20]. DeepSeek demonstrates that AI sovereignty is primarily an intellectual achievement, not a hardware or capital one. India’s response to chip export controls, or to any future geopolitical disruption in AI supply chains, will depend entirely on whether it has built the research base to innovate around constraints. Until now, it has not.

China also implemented an explicit institutional strategy to build presence in top AI conferences as a component of AI sovereignty. Universities such as Tsinghua, Peking, and Zhejiang were incentivised and funded to target publications at NeurIPS, ICML, and CVPR. Faculty hiring and promotion increasingly weighted top-venue publications highly. By 2023, Tsinghua University alone tied with Google for the most papers in the top-100 most-cited AI list, with eight each [19]. In comparison, no Indian institution appears in this ranking. The lesson here is that building R&D visibility at the world stage presence requires institutional strategy and incentive alignment, not merely researcher talent—which India possesses in abundance.

4.2. South Korea

South Korea’s R&D intensity, at 5.2% of GDP, is the highest in the world [17]. This investment transformed the country from a labour-intensive exporter of textiles and footwear in the 1960s to the global leader in semiconductors, displays, and batteries. The transformation was *deliberately* research-led: the Korea Institute of Science and Technology (KIST) and the Korea Advanced Institute of Science and Technology (KAIST) were founded (in 1966 and 1971, respectively) as national R&D anchors with explicit mandates for basic research. Samsung’s corporate R&D centre (SAIT) was restructured from a development-oriented laboratory to a basic-research institution to enable the firm’s transition from fast-follower to sovereign innovator [7]. In AI specifically, South Korea now leads the world in AI patents per capita at 17.3 per 1 Lakh inhabitants—a concrete measure of AI sovereignty in the intellectual-property dimension [19].

South Korea’s experience also illuminates the danger of the “fast-follower” model that India’s IT-services sector exemplifies. Korean policymakers recognised by the mid-1990s that the fast-follower strategy had reached its limits: as followers caught up, profitability was competed away, and the only escape was to move up the value chain through basic-research investment [7]. India’s IT sector faces the same structural pressure today, accelerated by rapid AI-driven commoditisation of software development and service-delivery tasks. The question is whether India will make the sustained, sovereignty-oriented research commitment South Korea made before its window of opportunity closes.

5. Recommendations for Building AI Sovereignty from the Research Up

The policy proposals below are sequenced to address the concerns identified in Section 3, rather than their symptoms. Together, they amount to a coherent programme for shifting India’s AI

investment from the surface layer to the foundational layer. The objective of our proposals should not be reduced to producing an Indian equivalent of today’s leading LLMs. The history of AI suggests that technological leadership can shift rapidly when a new paradigm emerges. Recent decades have seen successive transitions in the dominant AI paradigm, and the current foundation-model regime itself may eventually be displaced or complemented by approaches that are more data-efficient, interpretable, reliable, physically grounded, or computationally efficient. Potential areas include world models and physical intelligence, smaller and more controllable models, neurosymbolic approaches, novel learning paradigms, and new computational substrates such as analog or biological computing. The winners cannot be known *ex ante*, and that uncertainty is precisely why portfolio investment in fundamental research is necessary. India should therefore build current capabilities in foundation models while concentrating a smaller share of research resources on two or three plausible areas of future disruption. The objective is not to predict the next breakthrough, but to ensure that when the next paradigm arrives, India possesses a critical mass of researchers capable of contributing to it without going through another decade-long catch-up cycle.

5.1. Create a dedicated AI Research Fund

The Anusandhan National Research Foundation (ANRF), established in 2023, has been set up to foster impact-oriented research in the model of the United States National Science foundation (NSF). While the ANRF Advanced Research Grant (ARG) programme provides generous support for substantial R&D projects, and the ARG MATRICS programme supports basic research in mathematics, theoretical sciences, and quantitative social sciences, there is no explicit support for basic research in AI. We recommend that ANRF should earmark at least Rs.2,000 Cr over five years specifically for investigator-initiated, curiosity-driven AI research, encompassing learning theory, optimisation and algorithms, statistical machine learning, and the foundations of trustworthy AI. This fund should be modelled on the NSF’s Computing and Communication Foundations (CCF) programme, with rigorous international peer review at the level of NeurIPS and ICML programme committees, instead of UGC panels. No requirement for immediate application should be imposed. The objective of this programme would be to build the long-term intellectual foundations from which applied innovations and sovereign capability would emerge, while also creating sufficient flexibility to support research directions whose importance may become apparent only after a future technological shift. At approximately USD \$210 million over five years, the fund is a fraction of what a single frontier-class algorithmic innovation is worth in market terms, yet it would represent India’s first dedicated investment in foundational AI knowledge. A small fraction of such funding should explicitly support high-risk exploratory work outside the dominant foundation-model paradigm.

5.2. Reform academic incentives

UGC and AICTE promotion criteria in AI-relevant disciplines (e.g. Computer Science, Statistics, Data Science) should replace publication-count metrics with quality-weighted indicators. A

small number of genuinely high-quality contributions to internationally recognised research venues should carry substantially more weight in promotion and tenure decisions than large numbers of low-impact publications. Field-weighted citation impact and top-decile publication share should be incorporated into assessment frameworks. This single reform would redirect thousands of India’s AI researchers from volume-chasing to frontier-seeking, aligning individual incentives with the national AI sovereignty agenda. The cost to do so is negligible, but the potential impact is transformational.

5.3. Build autonomous frontier research organisations, not just data centres

India needs a small number of institutions explicitly designed to conduct high-risk, long-horizon AI research. The IndiaAI Innovation Centre could evolve toward a network of five to ten frontier research organisations, housed at institutions such as IISc, select IITs, ISI, CMI, TIFR, and other capable research institutions, but with substantially greater operational and financial autonomy than conventional government research programmes. Their mandate should be fundamental research rather than product delivery, with international peer review, multi-year funding, competitive recruitment, and freedom to pursue research directions whose applications may not be immediately visible.

To set up these frontier research organizations (FROs), the main constraint is their ability to attract and retain exceptional researchers. Where government salary rules and recruitment procedures prevent internationally competitive compensation, privately funded or public–private FROs can provide a complementary institutional model. Examples such as the Howard Hughes Medical Institute (HHMI), the Allen Institute for AI, Mila - Quebec Artificial Intelligence Institute, the Canadian Institute for Advanced Research (CIFAR), and other philanthropic research organisations demonstrate the potential of institutions that combine long-horizon scientific missions with operational independence. Following their operating model, protected endowments and governance structures that insulate researchers from short-term administrative pressures need to be created.

Such institutions would also provide a natural home for research on uncertain future directions—including world models, efficient and controllable AI, neurosymbolic approaches, and novel computing substrates—where conventional product incentives are poorly suited to funding long-horizon exploration. The objective is not to replace universities or public research agencies, but to create additional institutional capacity capable of operating at the frontier while broader reforms to India’s research ecosystem proceed.

5.4. Close the Indic data gap through research

The 1% representation of Indic languages in Common Crawl is not simply a data problem. It is a research problem in low-resource NLP, with interesting technical directions like cross-lingual transfer learning, code-switching modelling, and script-diverse tokenisation [10]. Funding data-centric AI projects as fundamental research programmes, rather than treating them solely as

dataset-creation exercises, would simultaneously address India’s linguistic AI gap and position India as a global leader in inclusive AI research. Sovereignty in AI for India requires AI that actually works for India’s population: this cannot be achieved by fine-tuning English-trained models, only by building the research foundations of multilingual intelligence.

5.5. Incentivise private-sector AI R&D at the frontier

India’s private sector contributes only 36% of national R&D expenditure, compared to over 70% in the United States, China, and South Korea [18]. Matched-funding schemes, in which government co-invests in corporate fundamental AI research rather than product development, should be established. Production-linked incentive (PLI) beneficiary firms in AI and electronics should be required to maintain a minimum R&D-to-revenue ratio and to publish at peer-reviewed venues. The long-term objective would be to cultivate a corporate research culture, at par with previously mentioned global standards such as AT&T Bell Labs and Samsung’s SAIT, that India currently lacks and that AI sovereignty ultimately requires.

To this end, policy should focus first on removing barriers rather than imposing publication or R&D quotas. Government can support industry–academic research through competitive grants, shared infrastructure, easier movement of researchers between universities and firms, and clearer rules for intellectual property and procurement. Where appropriate, forthcoming ANRF matching-funding mechanisms² can provide additional leverage for industry–academic collaborations.

The longer-term objective should be to create conditions under which Indian firms have an economic reason to establish genuine research laboratories rather than limiting AI activity to deployment and services. Public policy should therefore make it easier for firms to fund high-risk research, hire researchers competitively, collaborate with universities, and spin out technologies from research programmes.

5.6. Invest in research talent as a sovereignty asset

Talent is ultimately the binding constraint on every other component of AI sovereignty. India has a large population of technically trained graduates, but a much smaller pool of researchers operating consistently at the frontier of AI research. Moreover, this capacity is highly concentrated across a small number of institutions. No amount of compute or capital can compensate for an insufficient density of researchers capable of formulating and solving new problems.

In the short run, India should expand PhD and postdoctoral capacity in AI and related mathematical sciences, including joint PhD programmes with leading international universities and industrial PhD programmes connecting frontier researchers in industry with Indian institutions. Research-intensive institutions should also receive greater flexibility in recruitment, research staff appointments, and compensation, including mechanisms such as endowed chairs and

²https://anrfonline.in/ANRF/cpp_anrf

institutionally funded supplements where government salary rules remain binding.

In the longer run, institutional autonomy is essential. Leading Indian research institutions should have substantially greater authority over hiring, compensation, promotion, and research expenditure so that they can compete globally for exceptional scientists. The proposed FROs (Section 5.3) could serve as early experiments in this direction.

Finally, India should develop targeted programmes to attract researchers of Indian origin working at the global frontier. The objective should not simply be reverse migration, but the creation of enduring research networks linking Indian institutions with frontier laboratories and universities abroad. The central policy goal is to increase the density, not merely the number, of frontier researchers working from India.

6. Conclusion

India's current AI strategy risks producing a nation highly capable of adopting and deploying AI without building the research capacity required to shape the next technological frontier. The IndiaAI Mission, despite its ambition and scale, allocates zero rupees to fundamental AI research. Comparative evidence from China and South Korea demonstrates that bottom-up, sustained, quality-focused R&D investment is the common precursor to genuine technological sovereignty. China's GERD rose from 0.56 to 2.68% of GDP, and South Korea's rose from 1.68% to 5.21% over three decades whilst India's flatlined at 0.64%. Both these countries deliberately invested in basic research as a strategic national priority—and both now exercise a degree of AI sovereignty that India does not.

The concept of AI sovereignty goes beyond rhetoric and refers to concrete, measurable capabilities: the capacity to develop or adapt frontier AI under constrained conditions, the capacity to set the research agenda through high-impact publications, the capacity to generate foundational intellectual property rather than merely license it, and the capacity to build AI that serves India's population in its own languages and technological contexts, and the research depth to participate in future technological disruptions. India has a large base of technically trained human capital, strong research institutions, and ample strategic motivation to achieve this. What is missing is sufficient density of frontier researchers, institutional autonomy, and sustained funding for high-risk fundamental research. Recent initiatives such as the ANRF and the IndiaAI Mission provide important pieces of the infrastructure, but they need to be complemented by institutions capable of retaining exceptional researchers and pursuing questions beyond the immediate commercial frontier.

In this paper, we consciously focused on AI research instead of India's broader R&D landscape. That is because AI is a fast-moving, high-importance field dominated by a small number of countries and institutions in ways that make the sovereignty deficit especially acute and measurable. Nonetheless, the diagnoses advanced here apply with equal force to India's broader

scientific enterprise. AI is the domain where these pathologies are most visible and consequential today, but the underlying policy challenge is more general: India needs a research system capable not only of absorbing existing technologies, but also of generating original knowledge and positioning the country to participate in future technological disruptions. A research policy transformation that builds AI sovereignty would, by extension, begin to repair India's broader relationship with fundamental knowledge creation. The institutional reforms proposed here—quality-oriented incentives, greater autonomy for high-performing research institutions, stronger support for frontier talent, and new models of public and private research funding—are relevant well beyond AI. Thus, building AI sovereignty can serve as a test case for a broader transformation of India's research policy: from a system optimised primarily for publication volume and technology adoption towards one capable of sustained value creation that advances the frontiers of human knowledge. This would be consistent with the deeper *swadeshi* ethos of building indigenous capability rather than merely domesticising imported technologies.

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